Scan Matching

Given 2 point clouds, we want to match them up and find transformation b.w em, this tells us about where we are, how we've move, and is crucial for SLAM

1 Find how to calculate transformation assuming we know which points correspond

2 Find how to find which points correspo

1: Finding Transformation

Given inputs $X\{x_1 \dots x_j\}$, $Y\{y_1 \dots y_i\}$ w Correspondence $C\{(i,j)$

We want to find transformation t, rotation R such that

$$\sum_{(i,j)\in C} \| y_i - Rx_j - t \|^2 \text{ is minimized}$$

- $Rx + t = y$
  - known - found = error
  - $= y - (Rx + t) = y - Rx - t$

First, Simplify arrange so we use have pts. w corresp

$\{x_n\} \to \{y_n\}$, $\bar{x}_n = Rx_n + t$ for $n = 1 \dots |C|$

- Easier notation

3D we would also need $\bar{x}_n = \lambda R x_n + t$

Formal Defn

Given Points $\{(x_n, y_n)\}$ with possible certainty weights P_n

Find R, t st $\bar{x}_n = R x_n + t$

and $\sum \| y_n - \bar{x}_n \|^2 P_n$ is minimized

$\therefore$ Direct and Optimal Soln Exists
↳ no init guess

What this involves

1) Shift of COM 2) Rotation using SVD

Direct Soln for R

$$y_0 = \frac{\sum y_n P_n}{\sum P_n} \qquad x_0 = \frac{\sum x_n P_n}{\sum P_n}$$

Weighted Sum of Points

→ Cross Covariance Matrix $H = \sum (y_n - y_0)(x_n - x_0)^T P_n$

Outer Product now each pair of axes correlate

Then $SVD(H) = U D V^T$, Recall V = dirs in, U = dir out, D = scaling

$$R = V U^T$$

Direct Soln for T $= y_0 - R x_0$ simple!

$$\bar{x}_n = R(x_n - x_0) + y_0$$

To transform each point:

1) Take the variance of the source point from centroid
2) Rotate it as needed
3) Shift by centroid of target points

Derriving this Solution

- We want to use local coords set by Y_n

$\rightarrow$ Minimize $\sum \| Y_n - \underline{Y_0} - R X_n - t + \underline{Y_0} \|^2 P_n$ Does not change

- Rewrite translation vector

$\bar{X}_n = R X_n + t$

$\rightarrow \bar{X}_n - Y_0 = R X_n + t - Y_0$

$\rightarrow \bar{X}_n - Y_0 = R(X_n + \underbrace{R^T t - R^T Y_0})$ We can do since $R^T R = I$

Let's call this X_0, we don't know yet (turns out to be $\frac{\sum X_n P_n}{\sum P_n}$)

$\rightarrow X_n - Y_0 = R(X_n - X_0)$ where $= R^T t - R^T Y^0$

- So now

Find R, X_0 to minimize $\sum \| Y_n - Y_0 - R(X_n - X_0) \|^2 P_n$

- Obj func

$$\Phi(X_0, R) = \sum [(Y_n - Y_0) - R(X_n - X_0)]^T [(Y_n - Y_0) - R(X_n - X_0)] P_n$$

$\rightarrow$ Now minimize by finding derr etc

$$\Phi(x_0, R) = \sum (y_n - y_0)^T (y_n - y_0) p_n \quad \leftarrow \text{no } x_0, R$$

$$+ \sum (x_n - x_0)^T (x_n - x_0) p_n \quad \leftarrow \text{No } R \quad = \sum \| x_n - x_0 \|^2$$

$$- 2 \sum (y_n - y_0)^T R (x_n - x_0) p_n$$

① Solve WRT x_0

$$\frac{\partial \Phi}{\partial x_0} = -2 \sum (x_n - x_0) p_n + 2 \sum p_n (y_n - y_0)^T R \frac{\partial}{\partial x_0} (x_n - x_0)$$

$$\ldots \qquad - 2 \sum p_n (y_n - y_0)^T R$$

Lesson gradients must be col vectors, so check

$3\times1 - 3\times1$ ✓ $\quad 1\times3 - 1\times3$ NO, so T right side

$$= -2 \sum (x_n - x_0) p_n + 2 \sum R^T (y_n - y_0) p_n$$

- Set to 0 $\rightarrow \ldots \sum (x_n - x_0) p_n = R^T \underbrace{\sum (y_n - y_0) p_n}$

This must $=0$, we sum all points then subtract mean.
Mathematically, must $=0$

$$\rightarrow \sum (x_n - x_0) p_n = 0$$

From this, we find $\underline{x_0} = \dfrac{\sum x_n p_n}{\sum p_n}$ ✓ x_0 TURNS OUT to be this, we didn't set this

② Solve WRT R

Recall only $-2 \sum (y_n - y_0)^T R (x_n - x_0) p_n$ depends on R,

thus we need R to <u>maximize</u> $\sum (y_n - y_0)^T R (x_n - x_0) p_n$, with $R^T R = I$

Let $a_n = y_n - y_0$, $b_n = x_n - x_0$

$\rightarrow$ maximize $\sum a_n^T R b_n p_n$

We can rewrite using trace (By Defn of trace)

$\rightarrow$ max trace (RH) with $H = \sum (a_n^T b_n) p_n$ cross covar matrix

We'll use $SVD(H) = UDV^T$ where $U^TU = I$, $V^TV = I$, $D = \text{diag}$

Let's set $\underbrace{R = VU^T}$

since D Diag

$\rightarrow tr((VU^T)UDV^T) = tr(VDV^T) = tr(VD^{1/2}D^{1/2}V^T)$

$= tr((VD^{1/2})(D^{1/2}V)^T)$ Now let $A = VD^{1/2}$

$= tr(AA^T)$ with A being pos, def from SVD

Now we exploit inequality

For any pos def A, $tr(AA^T) \geq tr(R'AA^T)$ $\forall$ Rot R

From Schwarz Inequality

$\rightarrow tr(RH) = tr(AA^T) \geq tr(R'AA^T) = tr(\underbrace{R'R}H)$

This tells us RH is the maximum possible trace, As needed!

$\rightarrow R = VU^T$ was optimal!

- Any other R matrix is worse

Unique Soln? Only if Rank(H) = 3

Finally, $x_0 = R^T y_0 - R^T t \rightarrow t = y_0 - R x_0$

2: Finding Correspondences

As we might expect, no direct solution exists.
We must make an Iterative Soln: ICP

(Vanilla) Iterative Closest Point

Find transformation to align point clouds

① Find nearest neighbour

② Find transformation to do this
  ① Find COM and shift
  ② Find rotation using SVD method

Until Convergence

ICP Variants

1 Consider point subsets

2 Diff association strategies

3 Correspondence Weights

4 Outlier rejection

1 Point Subsets

- Uniform sub sampling
- Normal space sampling
- Feature based sampling
  - Get small set of highly distinct
  - corners, flowers, etc

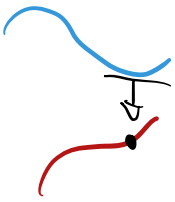
- good when flat planes, puts emphasis on surface features
- preprocessing
- Random Sampling

2 Data Associations

- Closest Point : baseline, for Vanilla ICP
  - Uses KD trees
  - stable but slow

  No init guess? Use COM of point clouds, match, then find closest

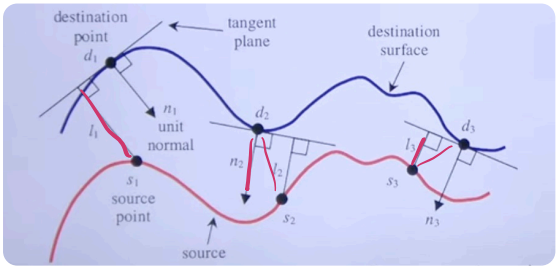
- Normal Shooting
  - Good for smooth



- Projective

- Point to Plane Popular

  source → closest point on plane of target



- Cost = (normal error) · (plane actual)
- Usually less iters needed

Point to Point

error

Point to Plane

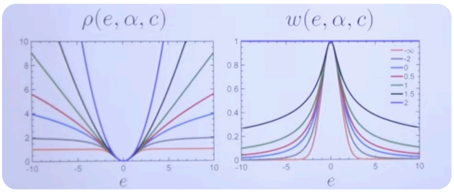
3 Weight Correspondences

Useful if known errors, ie stereo camera

4 Outlier Rejection

- Reject point to point dist > threshold
- Reject if neighbours don't agree / far away

Kernels



- Trim the worst __% correspondences
- Robust Kernel functions
- Heuristics important
  - What do we know?
- Important when Dynamic objects
  - We can identify and ignore em using residuals